THE AI TOOLBOX · CLASS 15

Research Showcase & Defense

学生科研展示与答辩

You present. The evidence speaks. The room asks what makes it trustworthy.

Live session · ≤ 6 slides including title · aim 4:30, hard stop 5:00 · 2:00 Q&A

HANDOFF · CLASS 14 → CLASS 15

Bring evidence, not a new experiment

Frozen from Class 14

- Question and bounded claim
- Data source, version and scope
- Saved split or folds + controls
- Pre-chosen metric and full results
- Notebook, limitation and next test

Today adds

- A six-slide evidence story
- A five-minute defense-ready talk
- Two teacher questions
- Glow · Grow · Next test feedback

Do not rerun until the answer looks better. If something changed after Class 14, disclose exactly what and why.

TODAY · THE ROOM CONTRACT

Students speak; the teacher tests the evidence

Presenter

Make one bounded claim, show the audit trail, stop on time, and say “I don't know” when the evidence ends.

Teacher

Ask at least two questions: one about **validity**, one about **interpretation or reproducibility**.

Peers

Listen for the chain of evidence and leave one Glow, one Grow and one Next test.

Privacy boundary: use a teacher-assigned presenter code. Show no full name, email, school, student ID, address, health detail, face, or raw personal record. Use aggregate or de-identified evidence only.

RUN OF SHOW · TWO-HOUR DEFAULT

One student = one eight-minute slot

Talk

aim 4:30 · hard stop 5:00

→

Teacher Q&A

2:00 · at least 2 questions

→

Transition

1:00 · feedback + next screen

Eight or fewer presenters

8 × 8 min = 64 min. A short preflight, opening, break and wrap all fit inside two hours.

More than eight

Split the showcase into separate sessions. Do not shrink talks, questions or feedback to force everyone into one room.

PREFLIGHT 1 OF 4 · SUPPORTED CLAIM

Quick check · tap an answer

A candidate model outperformed the baseline on the same predefined folds, but only one dataset was tested. Which headline is defensible?

- The candidate is best for every future dataset
- On this dataset and protocol, the candidate performed better; outside generalization is untested
- The experiment proves the model caused every improvement
- Remove the limitation so the conclusion sounds stronger

PREFLIGHT · THE SENTENCE TEST

Can the whole study fit in one honest sentence?

For **[population / dataset scope]**, we tested whether **[one change]** affected **[pre-chosen metric]** versus **[baseline]**, using **[saved protocol]**; we found **[bounded result]**, limited by **[largest threat]**.

Names the comparison

Change, baseline and outcome are visible.

Names the protocol

The audience knows how the test stayed fair.

Names the boundary

The claim stops where the evidence stops.

DECK RULE · MAXIMUM SIX SLIDES

Six slides, including the title

1 · Question
Title + bounded research question.

2 · Data
Source, version, n, scope and exclusions.

3 · Method
Change, baseline, controls and protocol.

4 · Results
Full comparison + uncertainty/error view.

5 · Meaning
Conclusion, limitations and rival explanation.

6 · Reproduce
Notebook/data links, AI disclosure, next test.

Six is a ceiling, not a target. Merge or cut anything that does not help the audience audit the claim.

STUDENT SLIDE 1 · QUESTION

Open with the decision the study can inform

Show

- Research title
- One precise, measurable question
- Presenter code — no identifying information

Say

“I tested whether ___ changes ___ for ___.” Then explain why that question matters in one sentence.

Do not begin with a biography, a table of contents, or five minutes of background.

STUDENT SLIDE 2 · DATA

Let the audience see what one row means

Provenance

Dataset name, source, version/date and license.

Scope

Sample size, unit of analysis, target and relevant balance/range.

Boundaries

Exclusions, missingness, likely bias and who/what is not represented.

Privacy: show aggregates, diagrams or de-identified examples. Never project raw student or participant records.

STUDENT SLIDE 3 · METHOD

Make the Class 14 contract visible

Baseline	→	One change	→	Controls	→	Metric
what must be beaten?		independent variable		same rows · folds · pipeline		chosen before results

Protocol details

Split/fold rule, seed, preprocessing, model settings and threshold rule.

Leakage defense

What was learned from training only? Were people, groups or future rows kept apart?

PREFLIGHT 2 OF 4 · FAIR TEST

Quick check · tap an answer

The candidate uses a new feature set and a new train/test split; the baseline uses the old features and old split. The candidate wins. What can the presenter defend?

- The new features caused the gain
- The candidate is universally better
- Only that the two completed runs differ; the cause is confounded
- The split never matters when the metric is identical

STUDENT SLIDE 4 · RESULTS

Show the whole comparison, not one winning number

Minimum evidence

- Baseline and every contender
- Pre-chosen metric
- Fold scores or interval/spread
- Confusion matrix or error plot when relevant

Keep visible

- Same evaluation protocol
- Units and denominators
- Null, failed or surprising result
- No cropped axis or cherry-picked subgroup

Evidence rule: every displayed number must be reproducible from the linked notebook. Placeholder or illustrative numbers must be labeled and cannot support the conclusion.

PREFLIGHT 3 OF 4 · HONEST HEADLINE

Quick check · tap an answer

A candidate catches more positives but also creates more false alarms than the baseline. Which headline is honest?

- The candidate is simply more accurate
- The candidate solves the task
- The candidate trades precision for recall; whether that helps depends on the stated error costs
- Hide precision because recall was the chosen metric

STUDENT SLIDE 5 · MEANING

Conclusion + limits belong on the same screen

Supported conclusion

Answer the question using the observed direction, size and uncertainty.

Largest limitation

Name the threat most likely to change the result — not a generic “more data.”

Rival explanation

What else could have produced the pattern, and what evidence would separate it?

Finish with the smallest claim that is still useful. “I cannot tell from this study” is a valid boundary.

STUDENT SLIDE 6 · REPRODUCE

Leave a route back to the evidence

Reproduction card

- Notebook/repository link
- Dataset identity and access instructions
- Environment/package versions
- Seed, split/fold and one-command rerun

AI disclosure

- Tool/model used
- What it helped produce
- What you checked, reran, corrected or rejected
- Or: “No generative AI used”

End with one next test that could challenge — not merely decorate — the conclusion.

BRIDGE · SLIDES ↔ SHORT PAPER

The same evidence, two reading speeds

Showcase slide	Paper section	What remains identical
1 · Question	Title + introduction	Scope and research question
2 · Data	Data / methods	Provenance, sample and exclusions
3 · Method	Methods / experiment	Baseline, controls and protocol
4 · Results	Results	Complete table, uncertainty and errors
5–6 · Meaning + reproduce	Discussion, limits, references	Bounded claim, caveat, artifacts and AI use

TALK MODEL · AIM 4:30

Six moves, with thirty seconds of safety margin

0:00–0:30
Question + why it matters

0:30–1:10
Data, source and scope

1:10–2:15
Design, controls and metric

2:15–3:30
Results + uncertainty/errors

3:30–4:10
Conclusion + largest limit

4:10–4:30
Reproduce + next test

Hard stop at 5:00. Rehearse to 4:30 so one pause, transition or clarification does not erase the ending.

DEFENSE · ANSWER STRUCTURE

Answer the question before adding context

1 · Listen

repeat the question if needed

→

2 · Answer

one direct sentence

→

3 · Evidence

point to result or protocol

→

4 · Boundary

what remains unknown

Useful starter

“From this experiment, I can say __ because __. I cannot yet tell __.”

If challenged

Thank the question, separate evidence from inference, and propose the next discriminating test.

PREFLIGHT 4 OF 4 · UNKNOWN ANSWER

Quick check · tap an answer

A teacher asks whether the result holds for a subgroup the project never analyzed. What is the strongest response?

- Yes, because the overall score was good
- Probably — then change the subject
- I do not know from this study; I would predefine the subgroup test and evaluate it without reusing the final test set
- Delete that subgroup from the dataset

TEACHER CARD · MINIMUM TWO QUESTIONS

One validity question + one meaning/reproduction question

Lane A · validity

- What changed, and what stayed fixed?
- How did you prevent leakage?
- Why does the split match real use?
- What baseline had to be beaten?
- Was the metric chosen before results?

Lane B · interpretation / reproduction

- What evidence would reverse your conclusion?
- Which limitation matters most?
- Can another student rerun this exactly?
- What AI output did you verify or reject?
- What is the next test, and why?

Ask at least one from each lane. Questions test ownership of the study, not trivia or public-speaking style.

PEER LISTENER CARD

Glow · Grow · Next test

Glow

The strongest **supported** claim or clearest evidence link.

Grow

One place where question → method → result → conclusion was unclear.

Next test

One feasible experiment that could challenge the claim or address its largest limit.

`“Glow: I trusted ___ because ___. Grow: clarify ___. Next test: hold ___ fixed and test ___.”`

ASSESSMENT · ALIGNED TO CLASS 14

Showcase + defense rubric · 100 points

Evidence category	Points	Full-credit signal
Question + bounded claim	**15**	Specific, measurable and no broader than the design
Data	**10**	Source/version/n/unit/exclusions and privacy-safe display
Experiment fairness	**20**	Credible baseline, one change, controls, saved split/folds and leakage defense
Results + uncertainty	**20**	Complete comparison, pre-chosen metric, spread/errors and no cherry-picking
Interpretation + limits	**15**	Evidence-based meaning, rival explanation, largest threat
Reproducibility + AI disclosure	**10**	Rerunnable artifacts and verified, specific AI-use record
Slides + delivery + timing	**5**	≤6 slides, readable, aim 4:30 and stop by 5:00
Defense / Q&A	**5**	Direct, evidence-linked answers with honest boundaries

Total: 100. A null result can earn full credit; unsupported certainty cannot.

PRESENTER · SIXTY-SECOND PREFLIGHT

Ready means the evidence can survive the projector

- Deck has **six slides or fewer**, title included
- Presenter code only; no student PII
- Every number comes from the final notebook
- Axes, units, denominators and labels are readable
- Limitation and next test are visible

- Notebook/data links open with correct permissions
- AI disclosure names verification actions
- Talk rehearsed to **4:30**
- One validity answer prepared
- One interpretation/reproduction answer prepared

If a screen contains private data, close it before sharing. A screenshot is not a privacy plan. Open printable showcase worksheet ↗

LIVE CONTROL · PRESENTER

4:30 aim

0:00	→	1:00	→	2:15	→	3:30	→	4:30
question		method		results		meaning		finish

Hard stop: 5:00. At 4:30, finish the current sentence and land the limitation/next test.

LIVE CONTROL · TEACHER DEFENSE

2:00 Q&A

Question 1 · validity

Design, baseline, controls, leakage, split or metric.

Question 2 · interpretation / reproduce

Claim boundary, limitation, rerun, AI verification or next test.

Two questions are the minimum. Keep answers direct so both lanes are tested.

LIVE CONTROL · TRANSITION

1:00 feedback + switch

Peers

Finish Glow · Grow · Next test.

Teacher

Capture rubric evidence; do not write from memory later.

Next presenter

Open deck, confirm presenter code, start on title slide.

No public ranking. Return feedback privately or through the class's approved channel.

AFTER THE DEFENSE · FINAL PACKAGE

Submit what another student needs to rerun

Required

- Final ≤6-slide deck
- Short paper or structured report
- Runnable notebook/repository
- Dataset citation + access instructions
- Environment, seed and split/fold record

Final audit

- Results agree across paper, slides and code
- Limitation and next test are consistent
- AI-use disclosure is specific
- Private data and student PII are removed
- Links work for the intended reviewer

Do not submit raw personal records. Use the course's approved private workflow for any sensitive source data; public artifacts contain aggregates or de-identified derivatives only.

WRAP-UP · THE RESEARCH LOOP CLOSES

A defense is not certainty.
It is visible ownership of the evidence.

Claim

Small enough for the design to support.

Chain of custody

Question → data → protocol → result → limit.

Next test

Specific enough to challenge what you believe.

Exit line: “The strongest thing my evidence supports is ___; the next test is ___.”

More than eight presenters? Continue in a separate full-length showcase session.